

Bitwise linear trail through **LBlock** found by SAT

CiVerLy

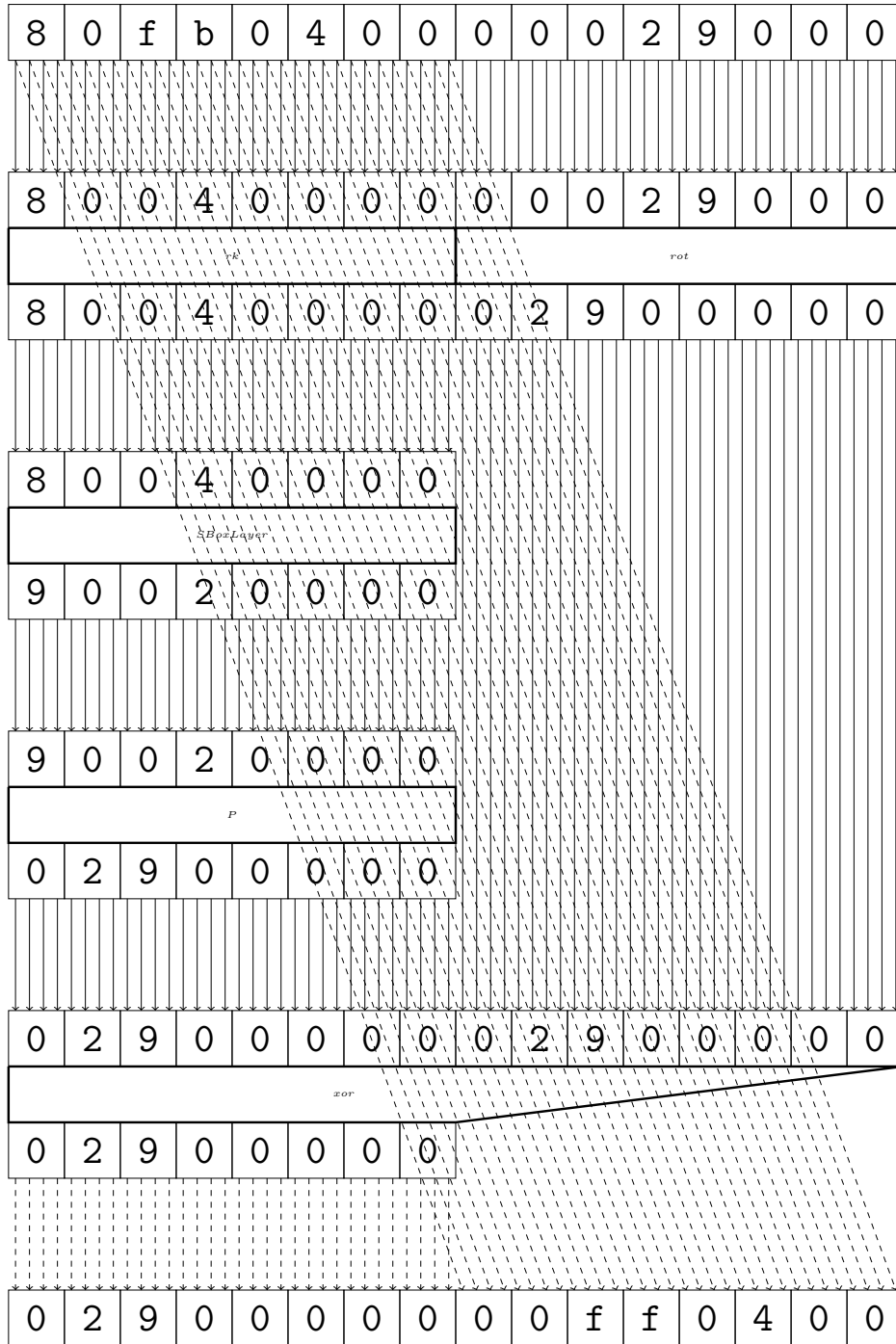
July 20, 2026

1 LBlock

Maximal linear correlation: 2^{-33}

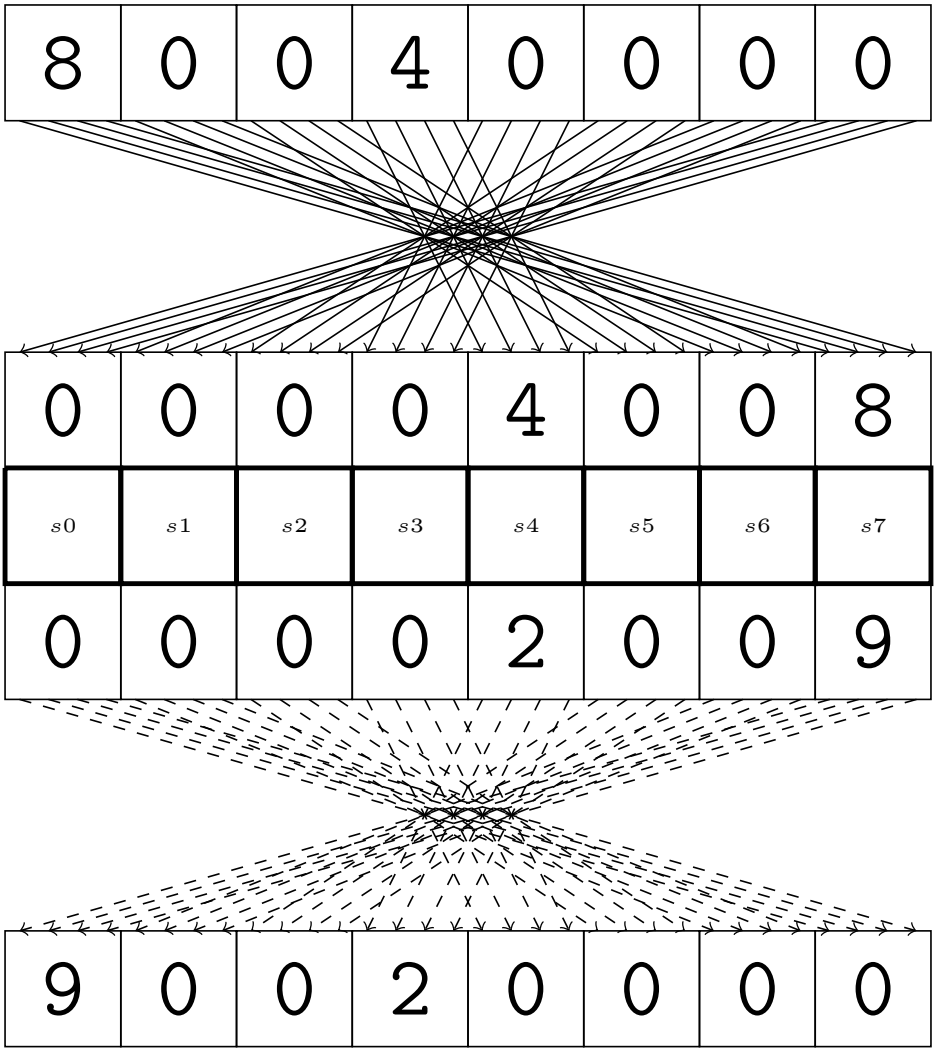
8	0	f	b	0	4	0	0	0	0	2	9	0	0	0
8	0	f	b	0	4	0	0	0	0	2	9	0	0	0
0	2	9	0	0	0	0	0	0	0	f	f	0	4	0
0	2	9	0	0	0	0	0	0	0	f	f	0	4	0
f	f	0	4	0	0	0	0	0	0	0	2	0	0	0
f	f	0	4	0	0	0	0	0	0	0	2	0	0	0
0	2	0	0	0	0	0	0	f	f	0	0	0	0	0
0	2	0	0	0	0	0	0	f	f	0	0	0	0	0
0	0	0	0	0	0	f	f	0	2	0	0	2	0	2
0	0	0	0	0	0	f	f	0	2	0	0	2	0	2
0	0	2	0	2	0	0	2	f	0	0	0	f	0	f
0	0	2	0	2	0	0	2	f	0	0	0	f	0	f
0	0	0	f	0	f	f	0	0	0	0	0	0	0	0
0	0	0	f	0	f	f	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	f	0	f	f
0	0	0	0	0	0	0	0	0	0	0	f	0	f	f
0	f	0	f	f	0	0	0	0	2	c	0	c	0	0
0	f	0	f	f	0	0	0	0	2	c	0	c	0	0
2	c	0	c	0	0	0	0	0	5	7	f	0	0	0
2	c	0	c	0	0	0	0	0	5	7	f	0	0	0
5	7	f	0	0	0	0	0	0	0	0	0	0	0	0
5	7	f	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	5	7	f	0	0	0	0	0
0	0	0	0	0	0	0	5	7	f	0	0	0	0	0
f	0	0	0	0	5	7	0	b	0	0	2	0	3	0
f	0	0	0	0	5	7	0	b	0	0	2	0	3	0
0	0	2	0	3	0	0	b	0	0	0	0	f	0	7
0	0	2	0	3	0	0	b	0	0	0	0	f	0	7
0	0	0	f	0	7	0	0	0	0	0	3	0	0	0
0	0	0	f	0	7	0	0	0	0	0	3	0	0	0
0	0	3	0	0	0	0	2	0	0	f	0	7	0	0
0	0	3	0	0	0	0	2	0	0	f	0	7	0	0
2	0	0	f	0	7	0	0	0	0	3	0	0	0	0
2	0	0	f	0	7	0	0	0	0	3	0	0	0	0

2 LBlock_round

Maximal linear correlation: $2^{-2.0}$ 

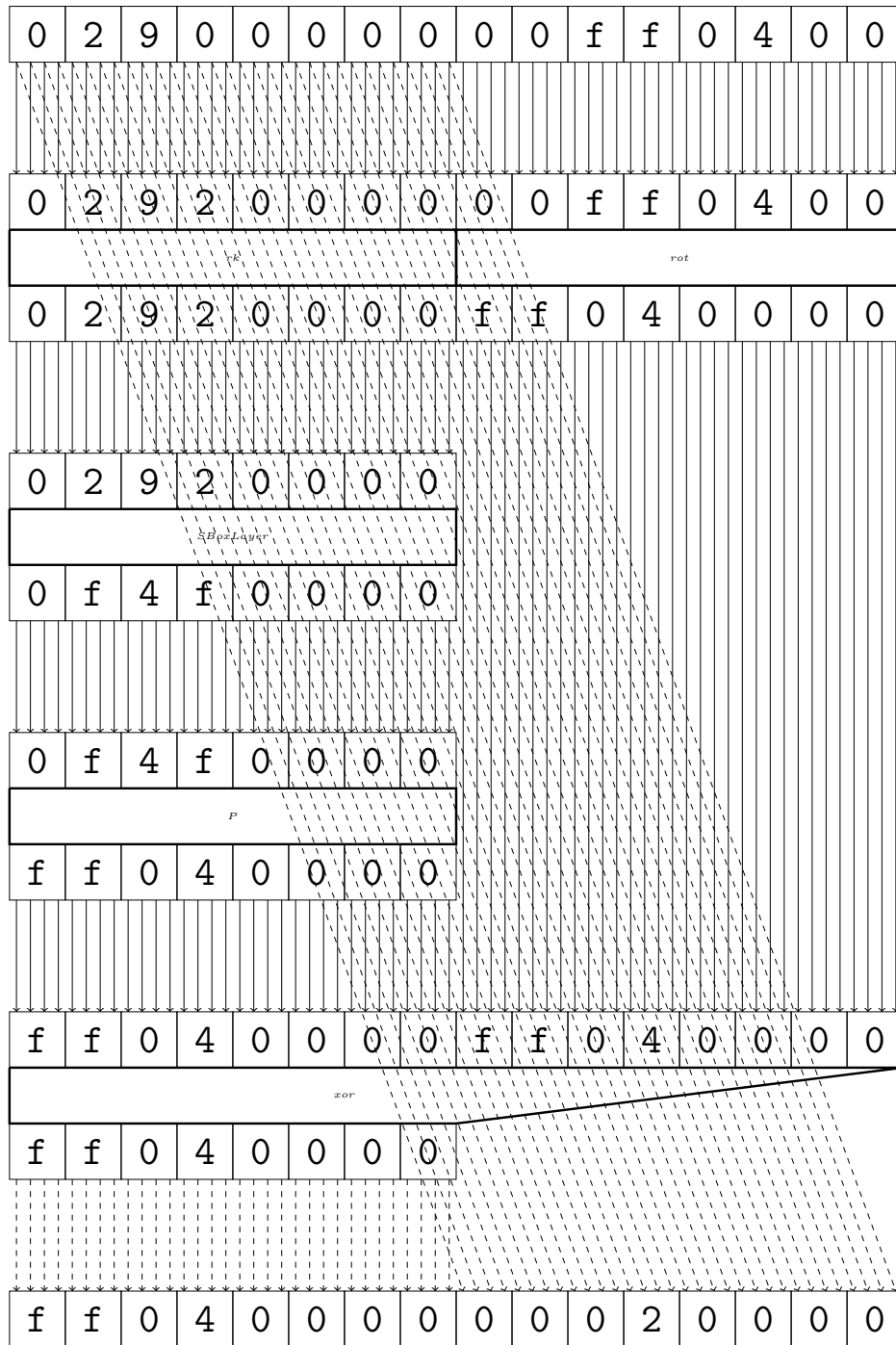
3 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



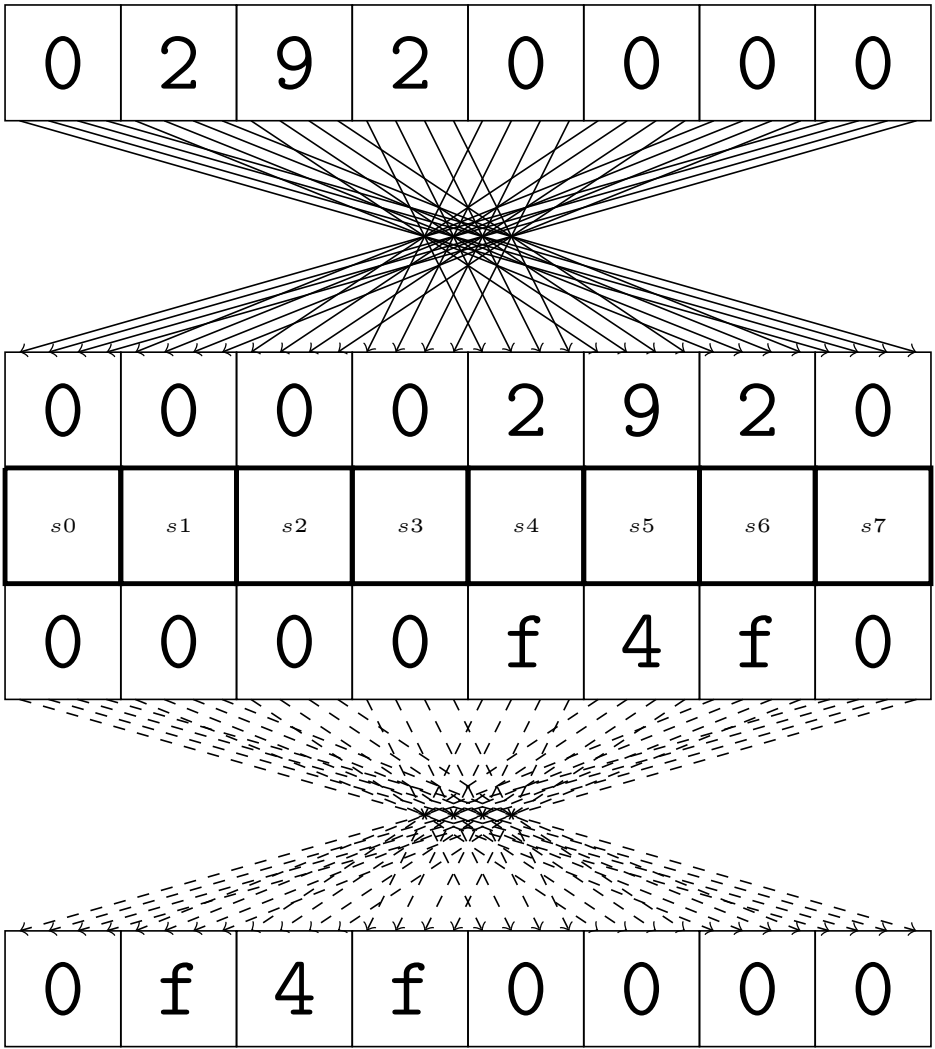
4 LBlock_round

Maximal linear correlation: $2^{-3.0}$

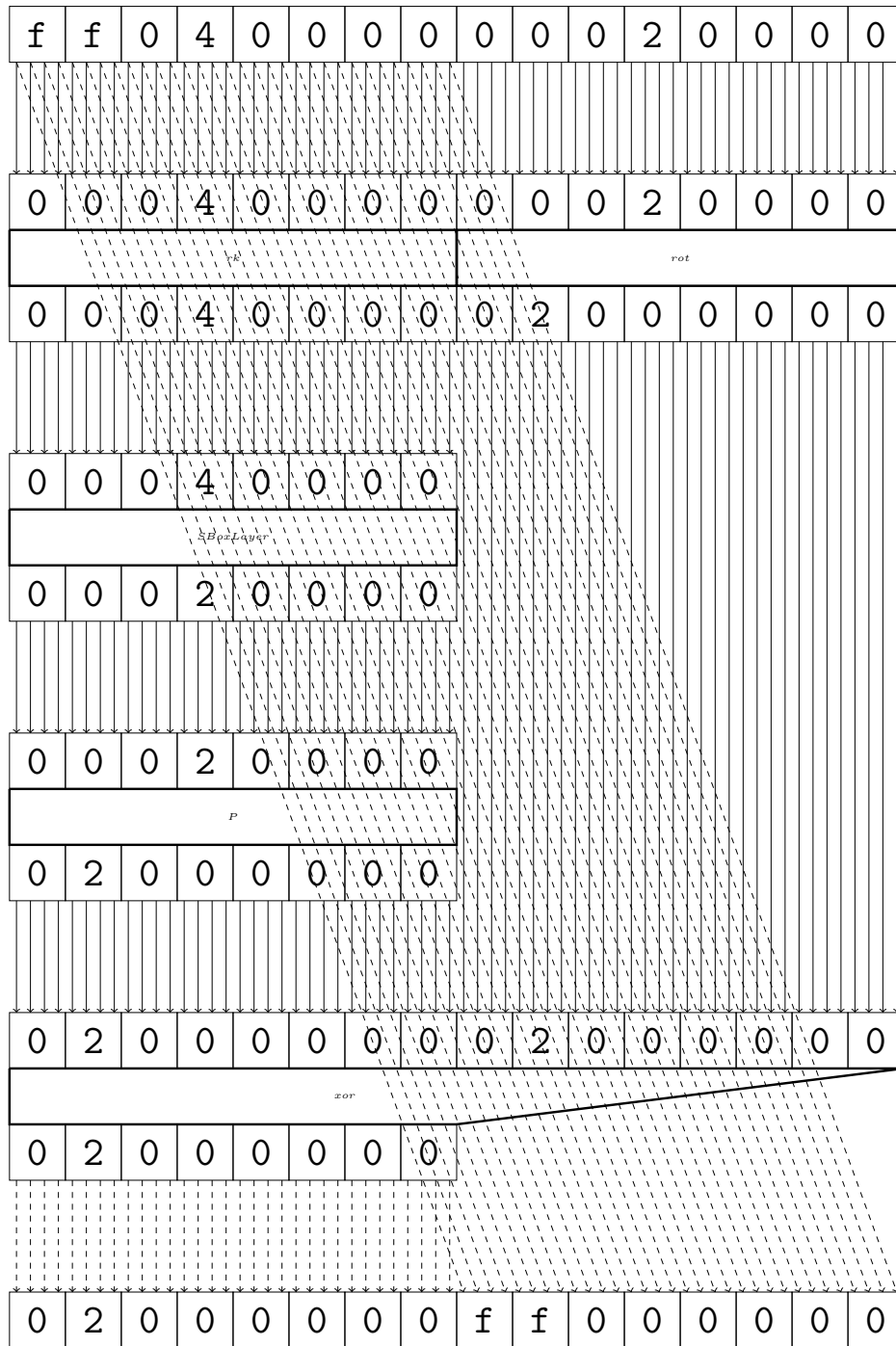


5 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

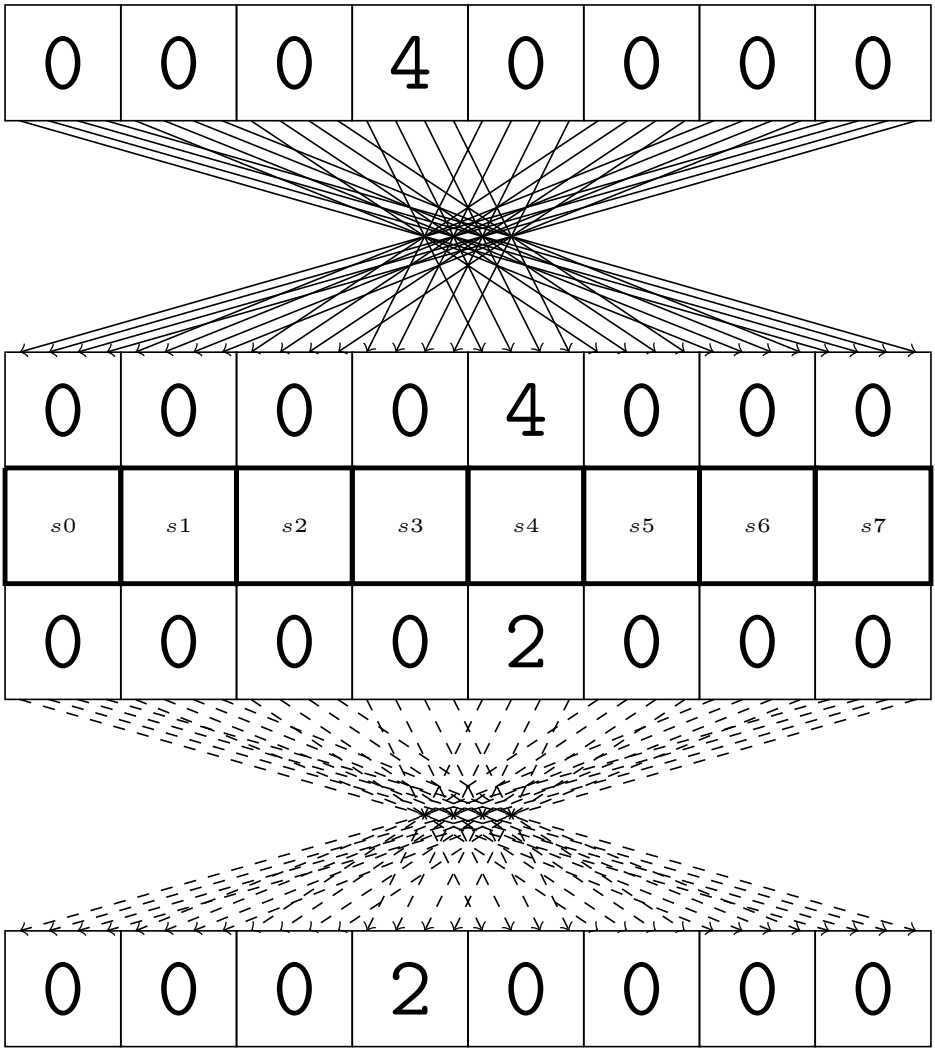


6 LBlock_round

Maximal linear correlation: $2^{-1.0}$ 

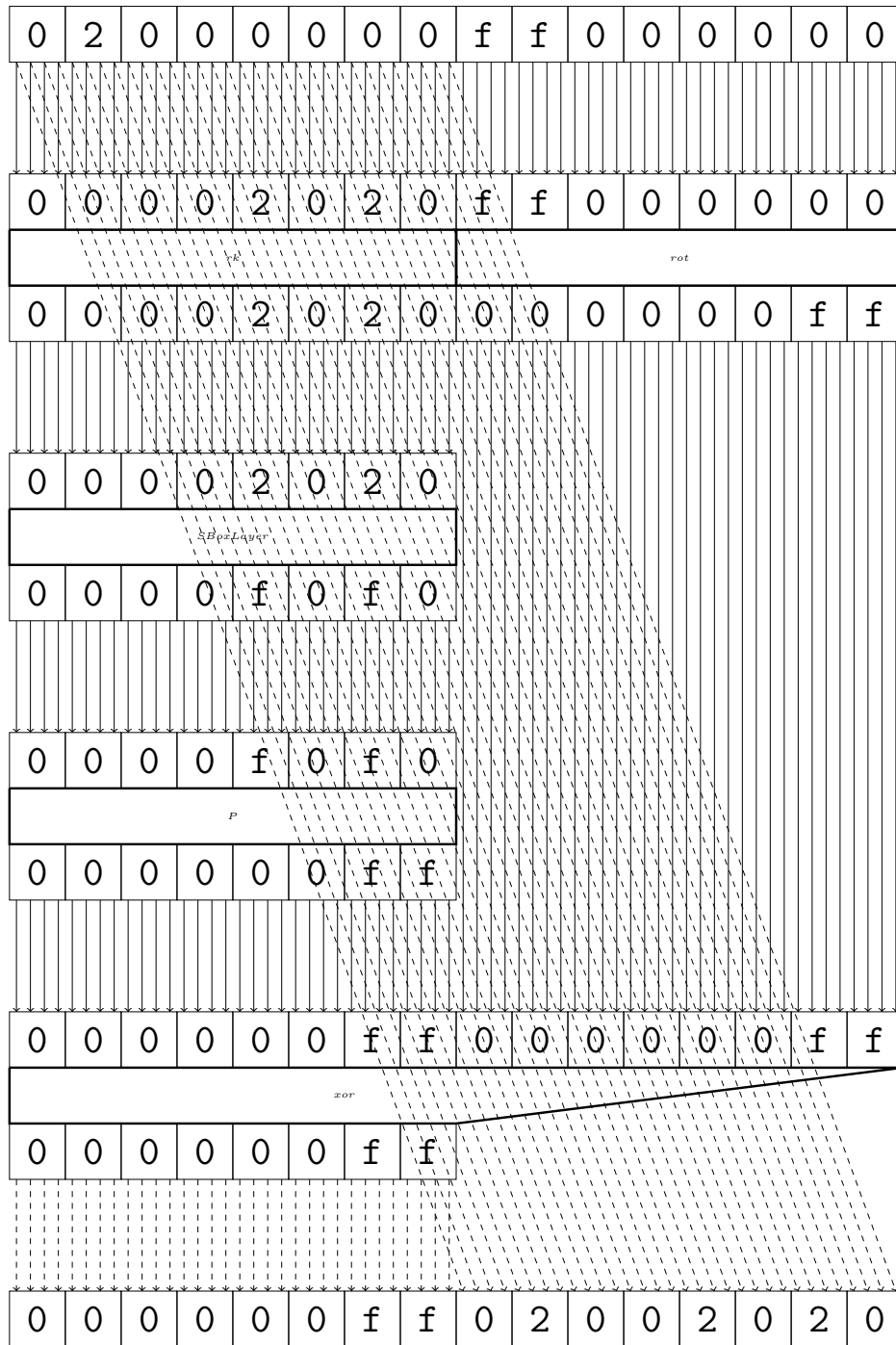
7 SBoxLayer

Maximal linear correlation: $2^{-1.0}$



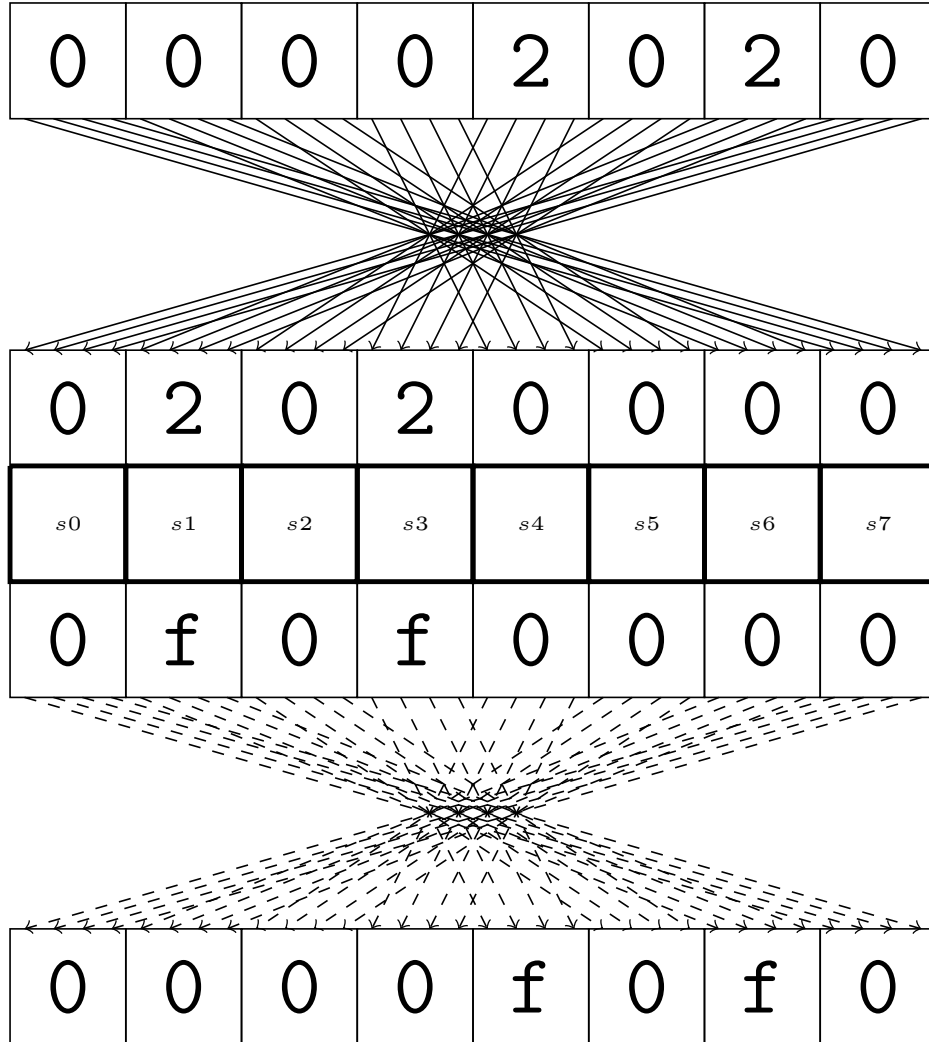
8 LBlock_round

Maximal linear correlation: $2^{-2.0}$



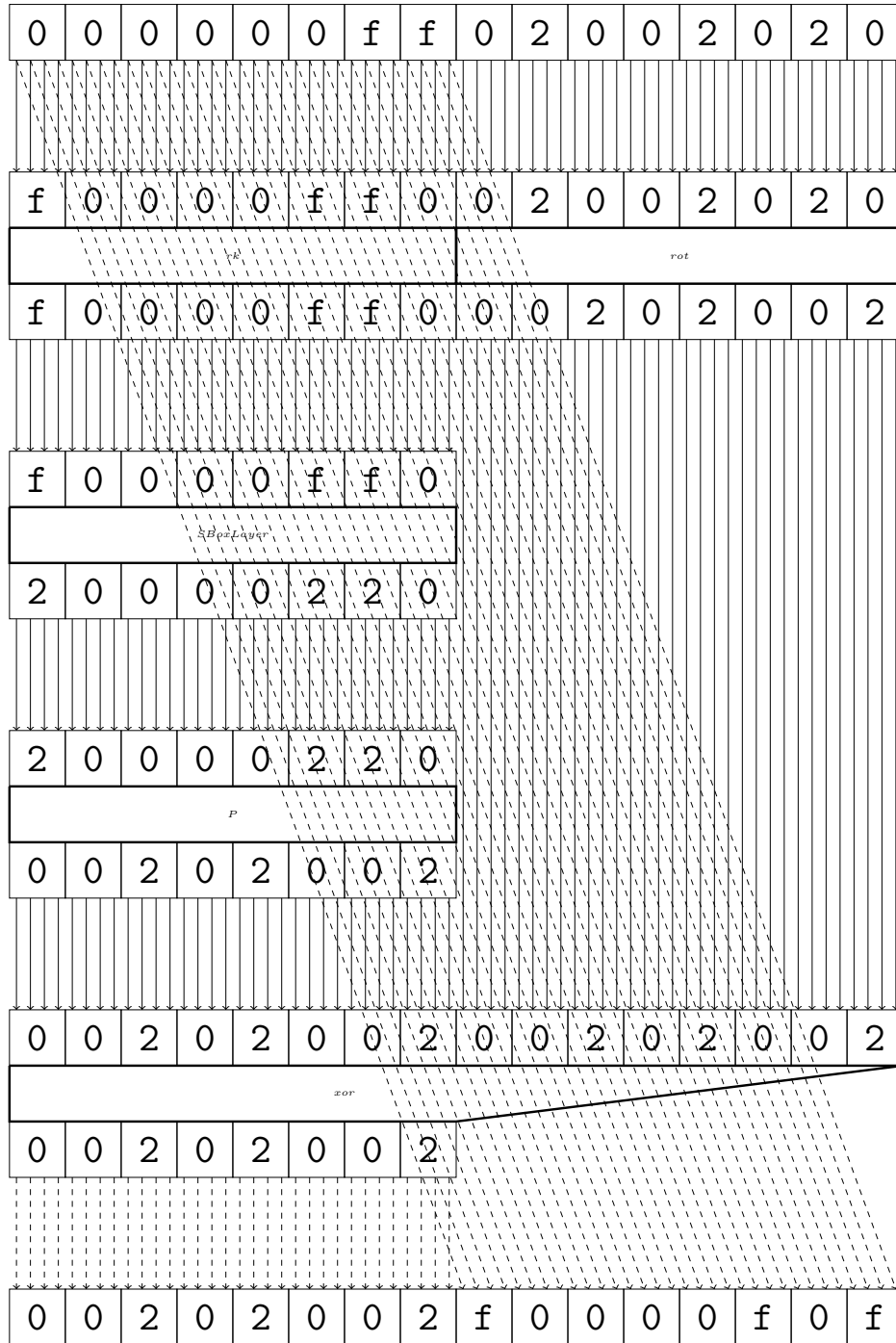
9 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



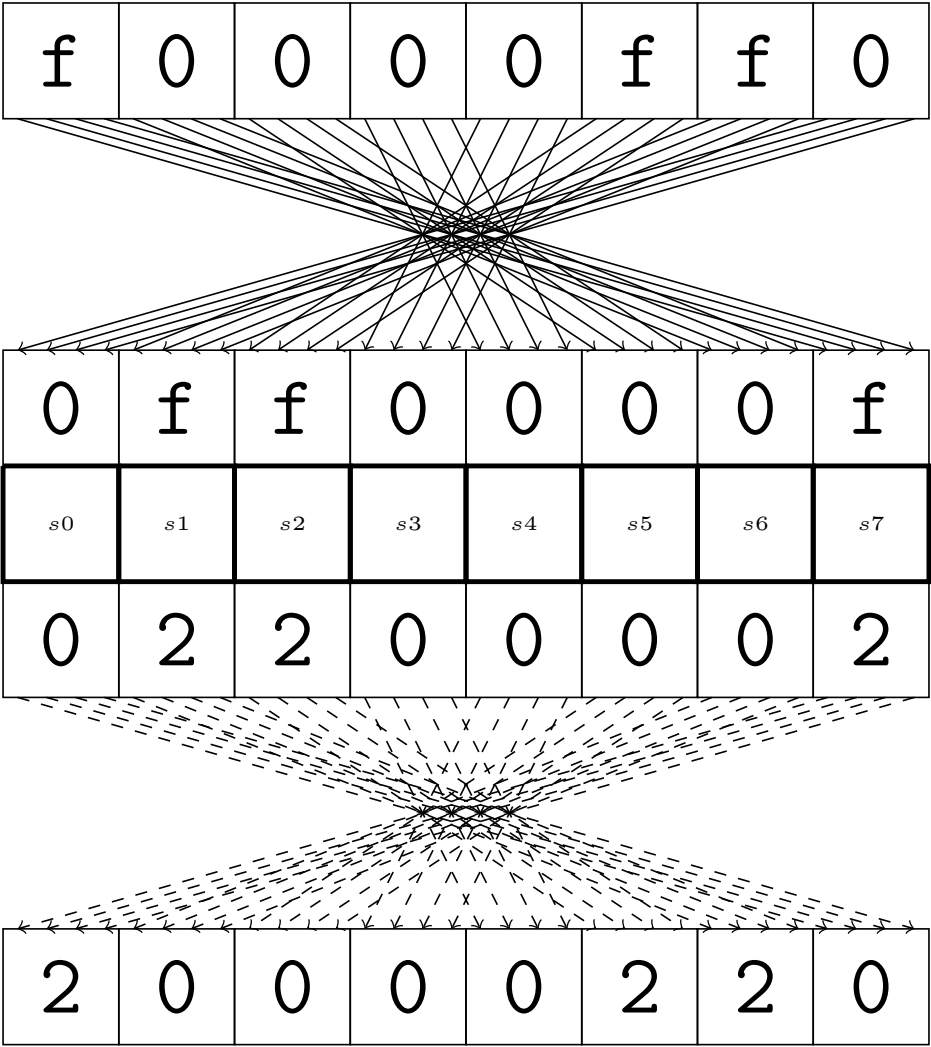
10 LBlock_round

Maximal linear correlation: $2^{-3.0}$



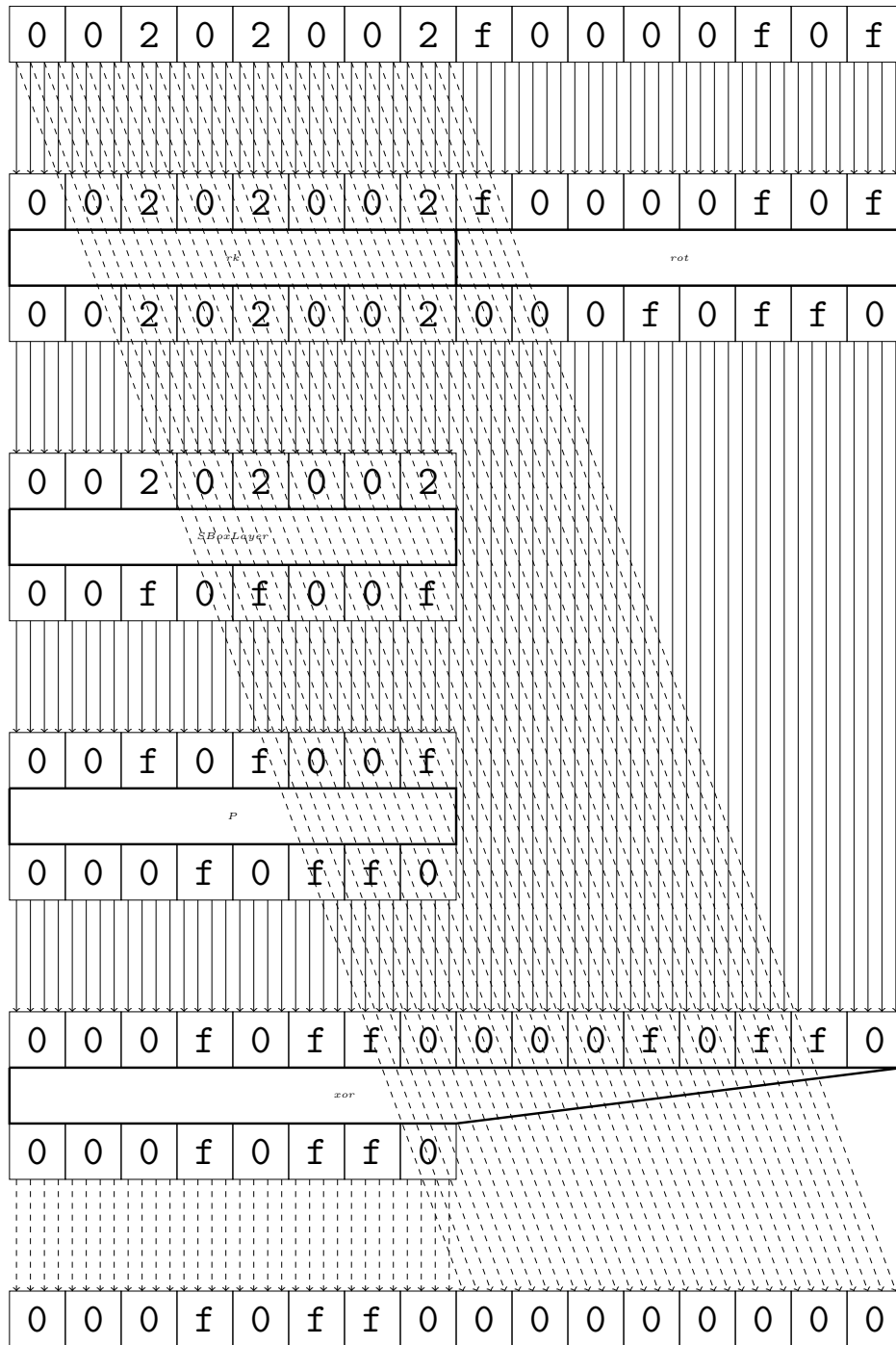
11 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



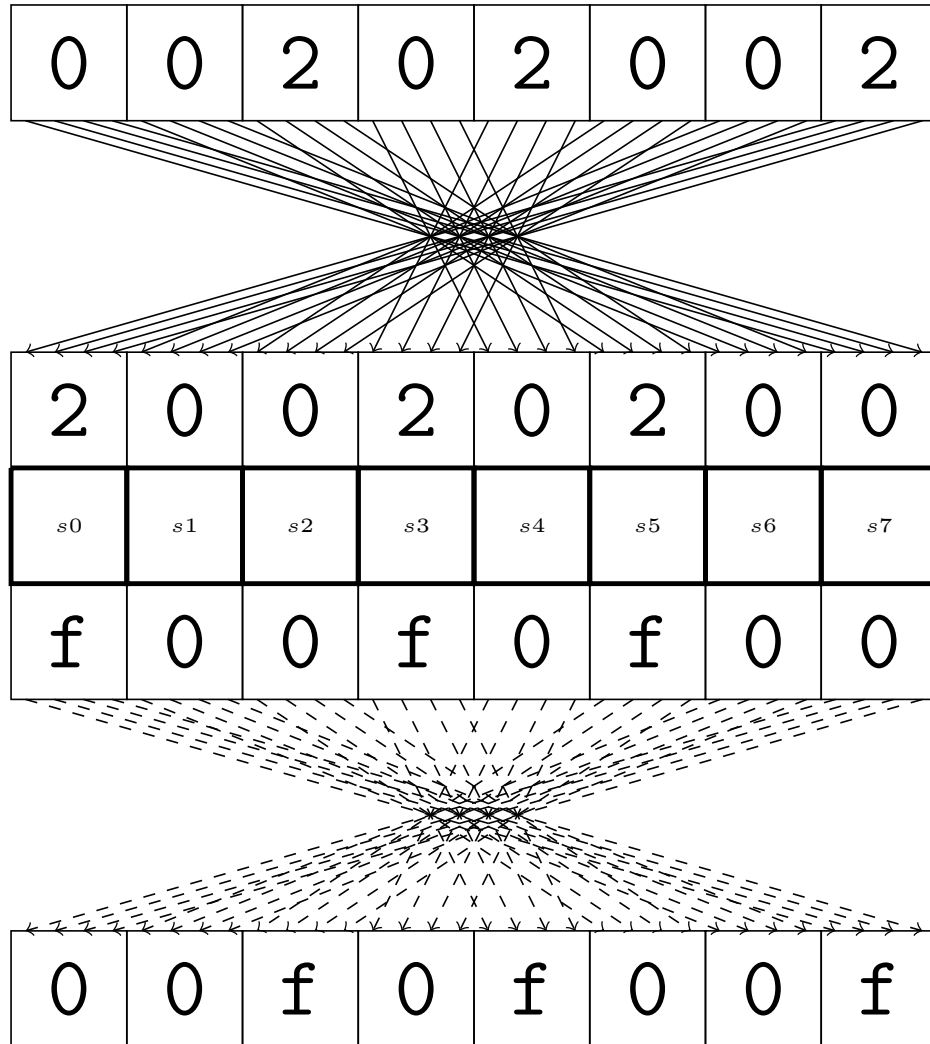
12 LBlock_round

Maximal linear correlation: $2^{-3.0}$

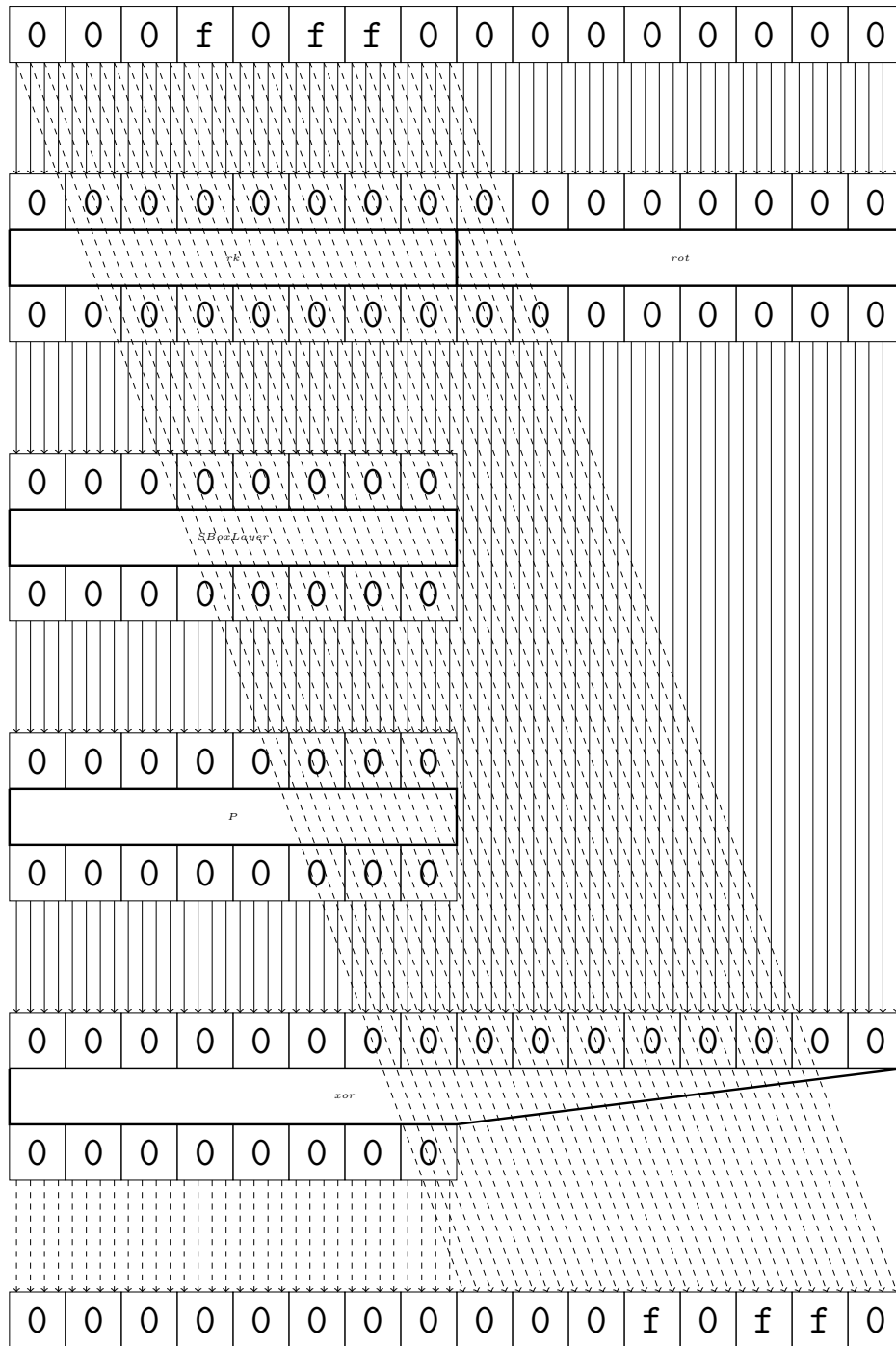


13 SBoxLayer

Maximal linear correlation: $2^{-3.0}$

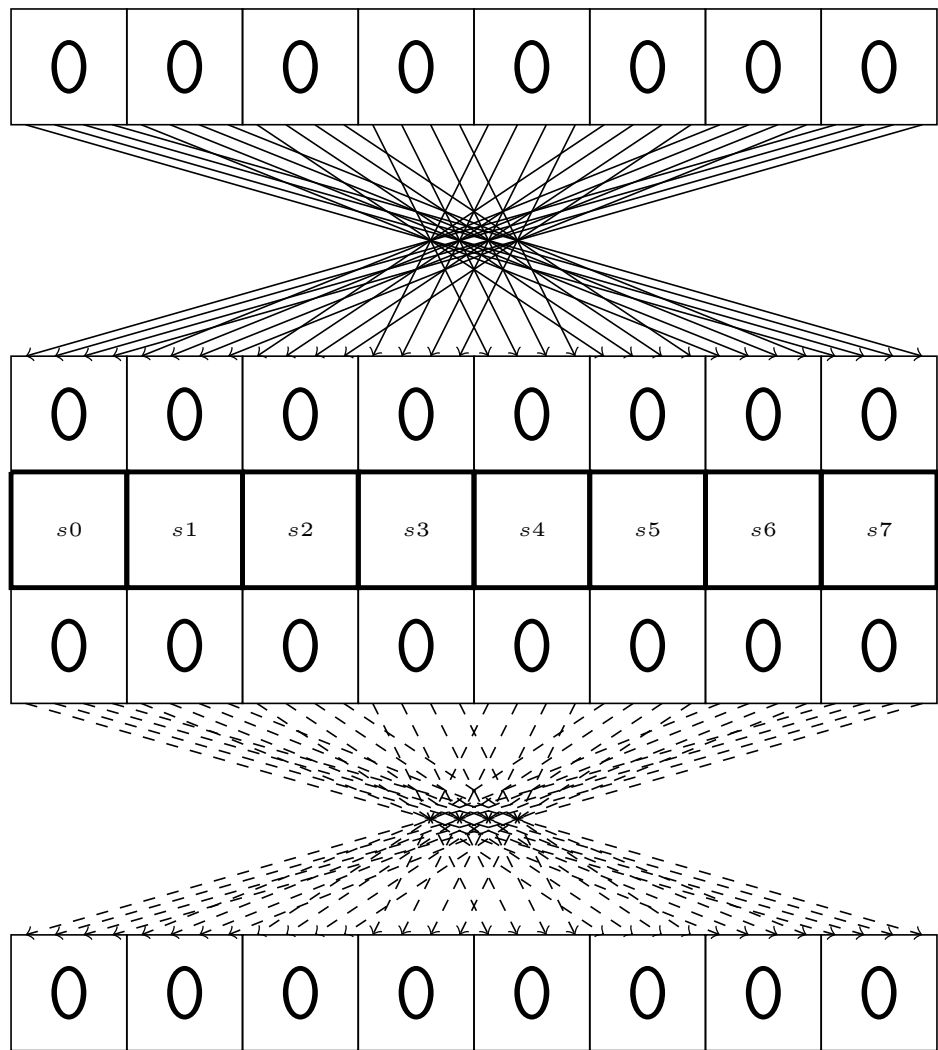


14 LBlock_round

Maximal linear correlation: $2^{-0.0}$ 

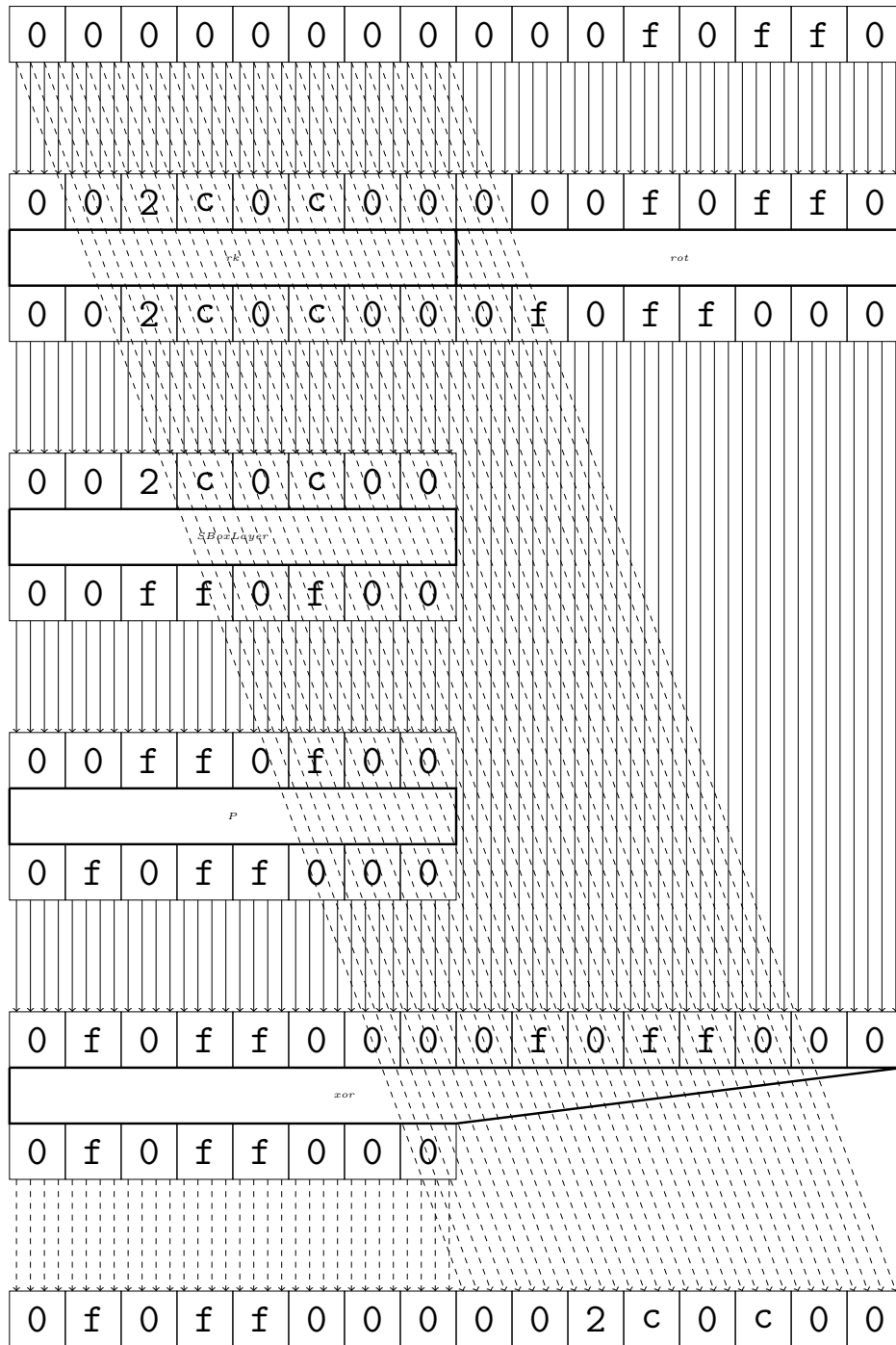
15 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



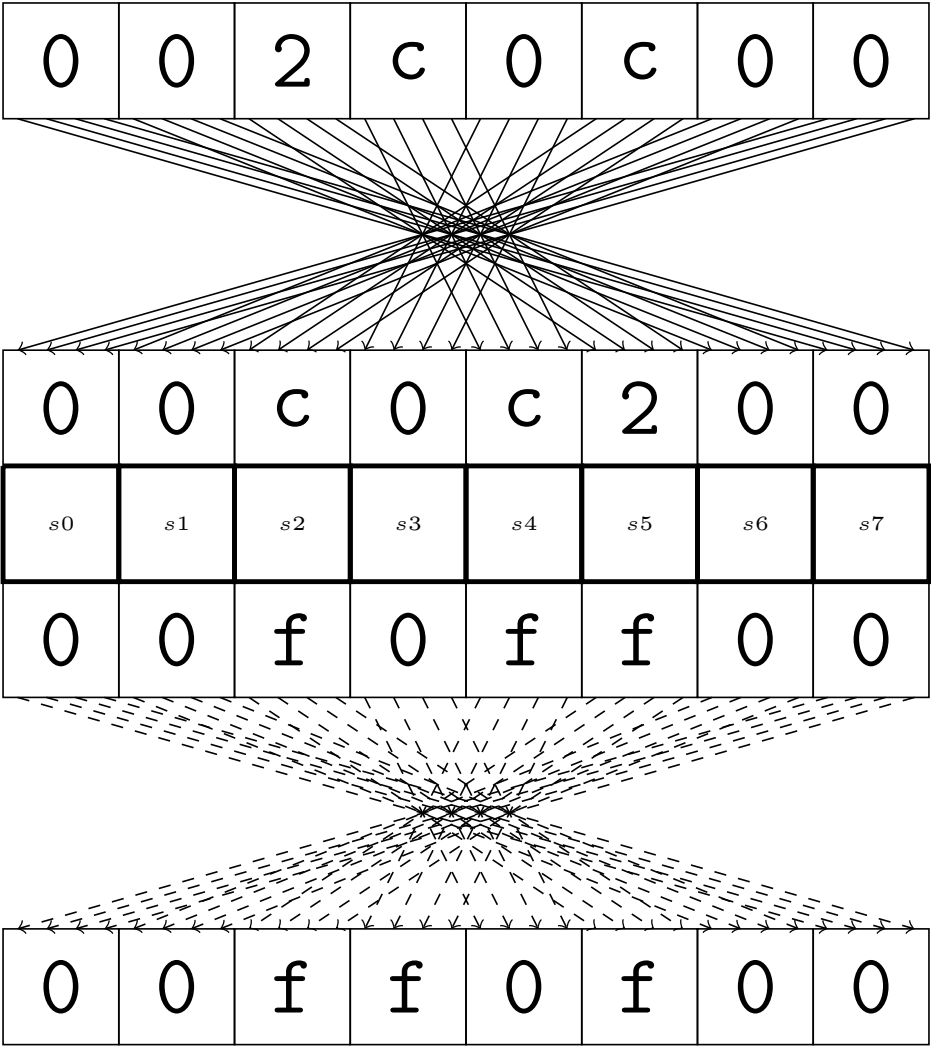
16 LBlock_round

Maximal linear correlation: $2^{-3.0}$



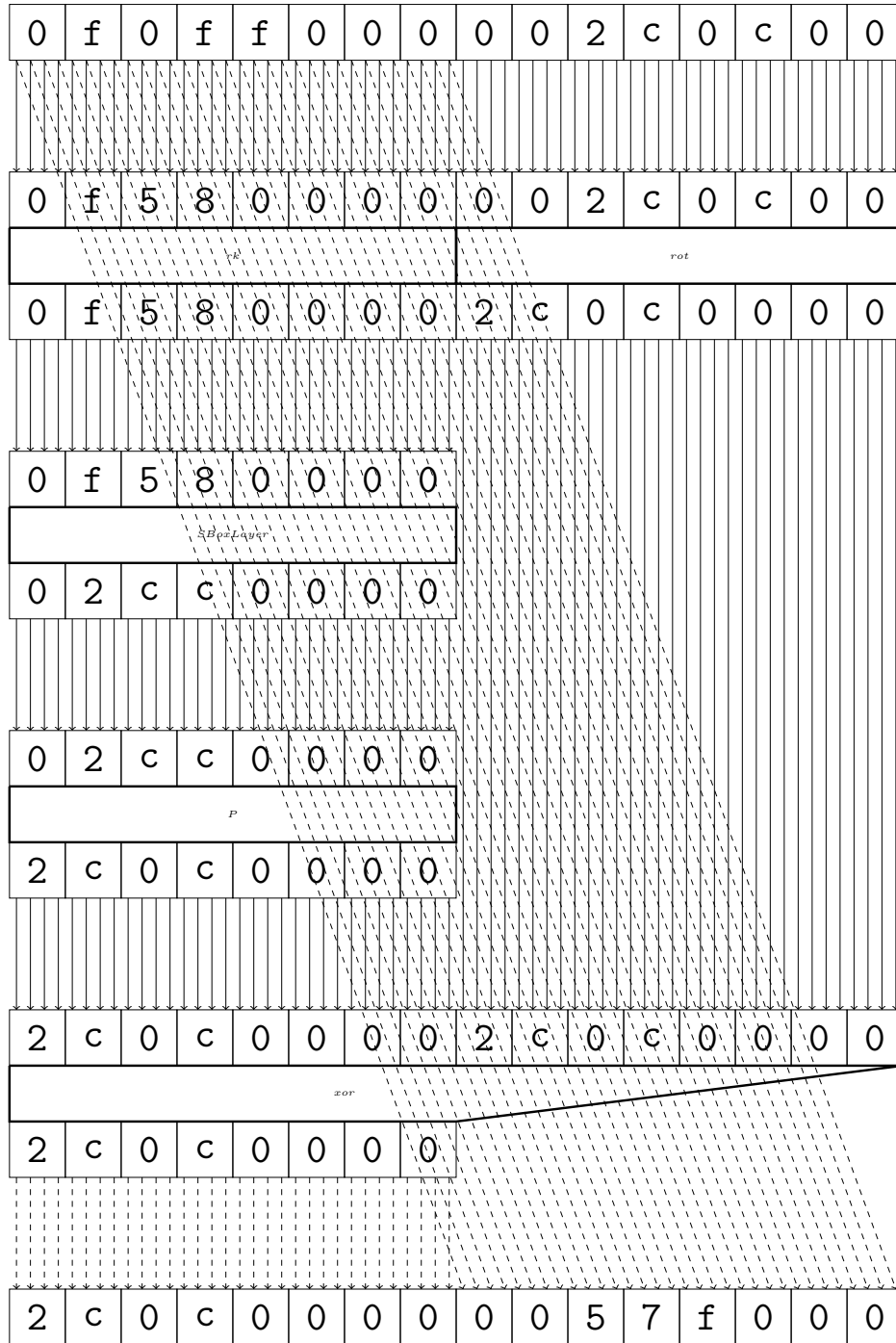
17 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



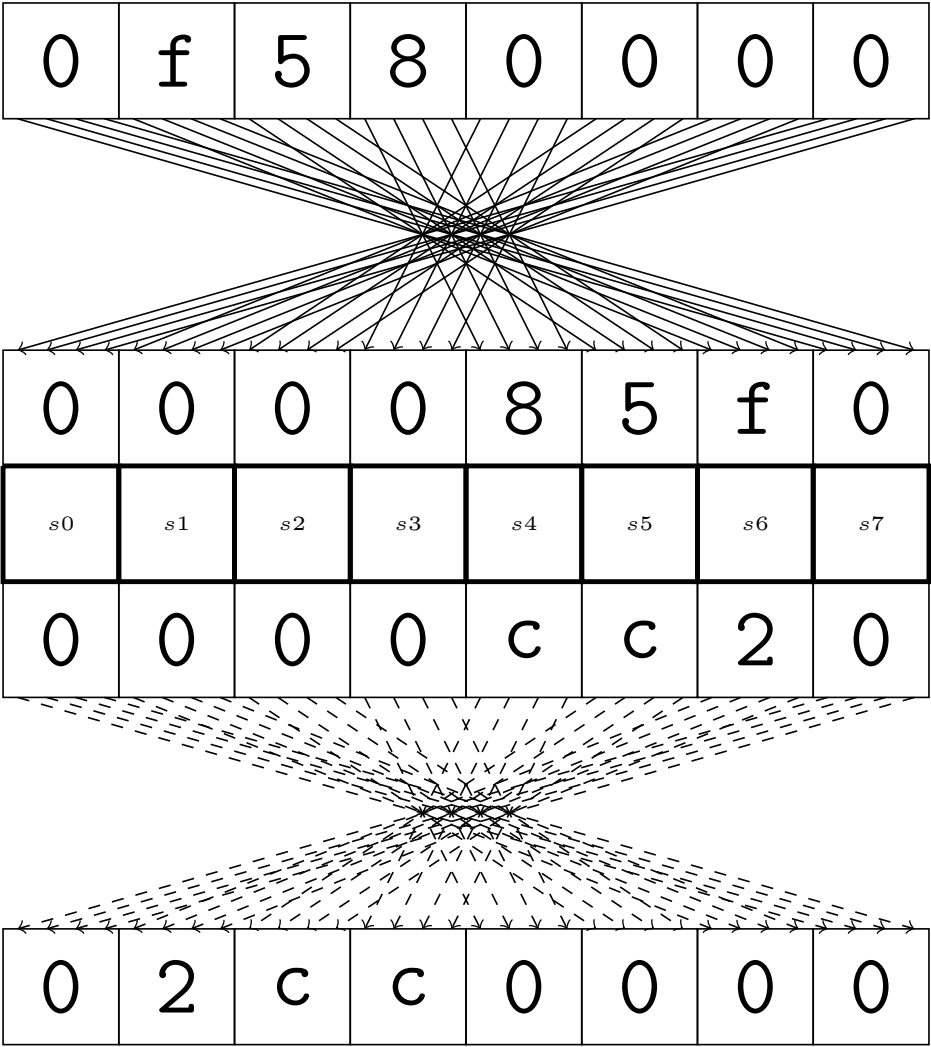
18 LBlock_round

Maximal linear correlation: $2^{-3.0}$



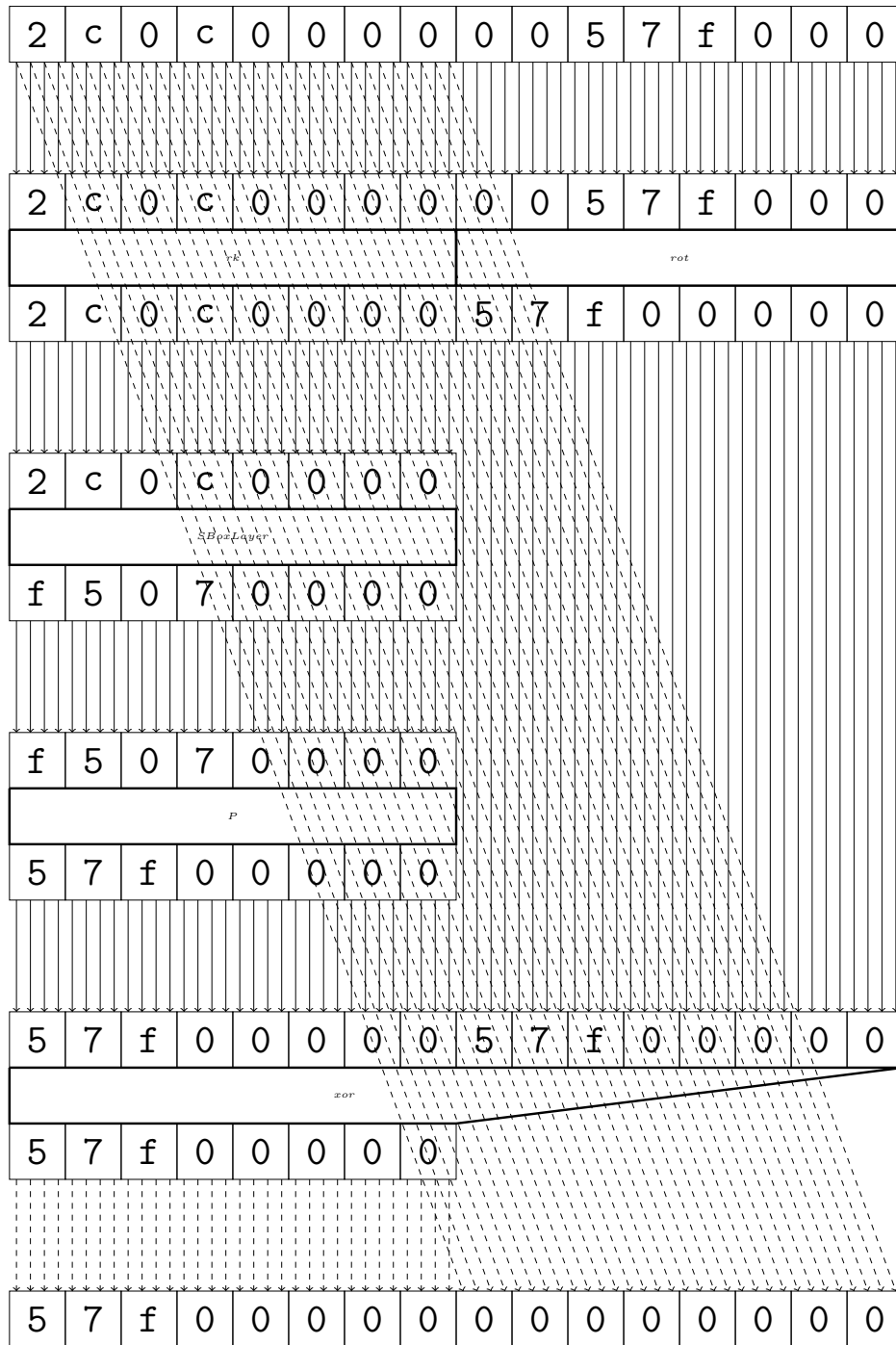
19 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



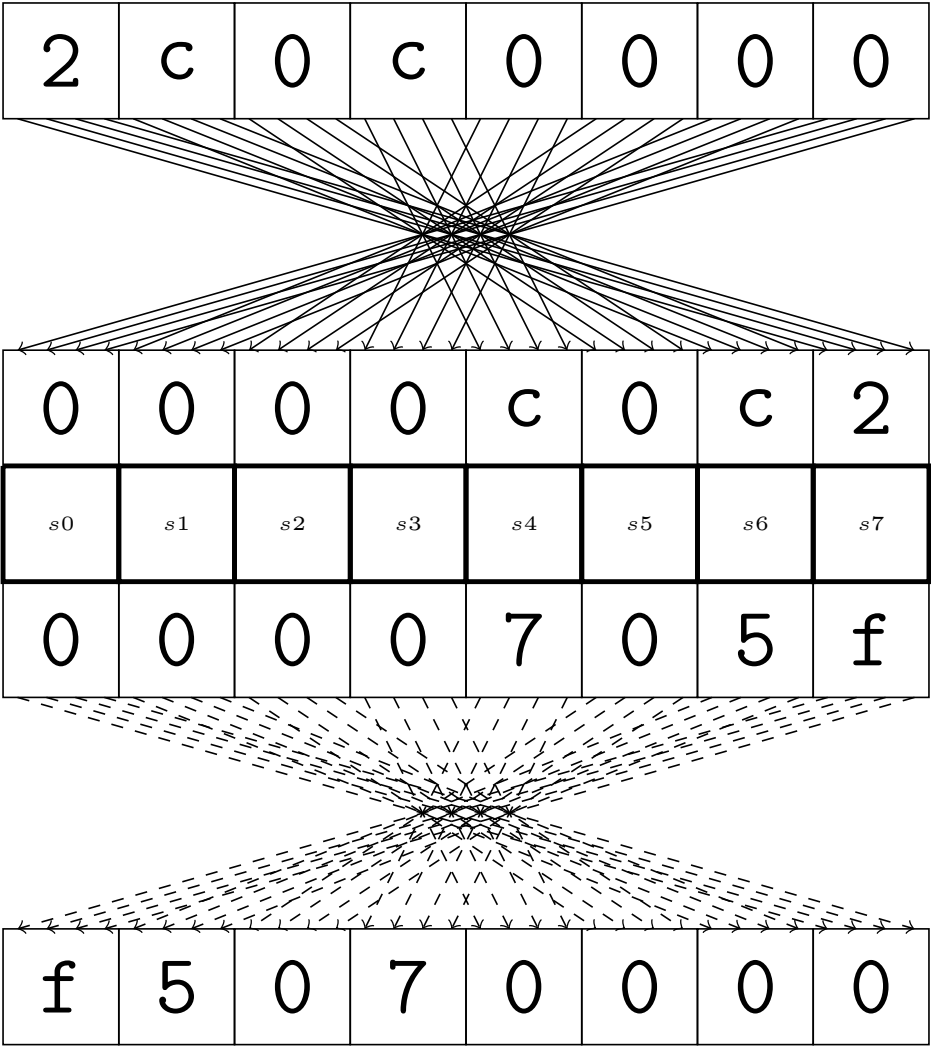
20 LBlock_round

Maximal linear correlation: $2^{-3.0}$



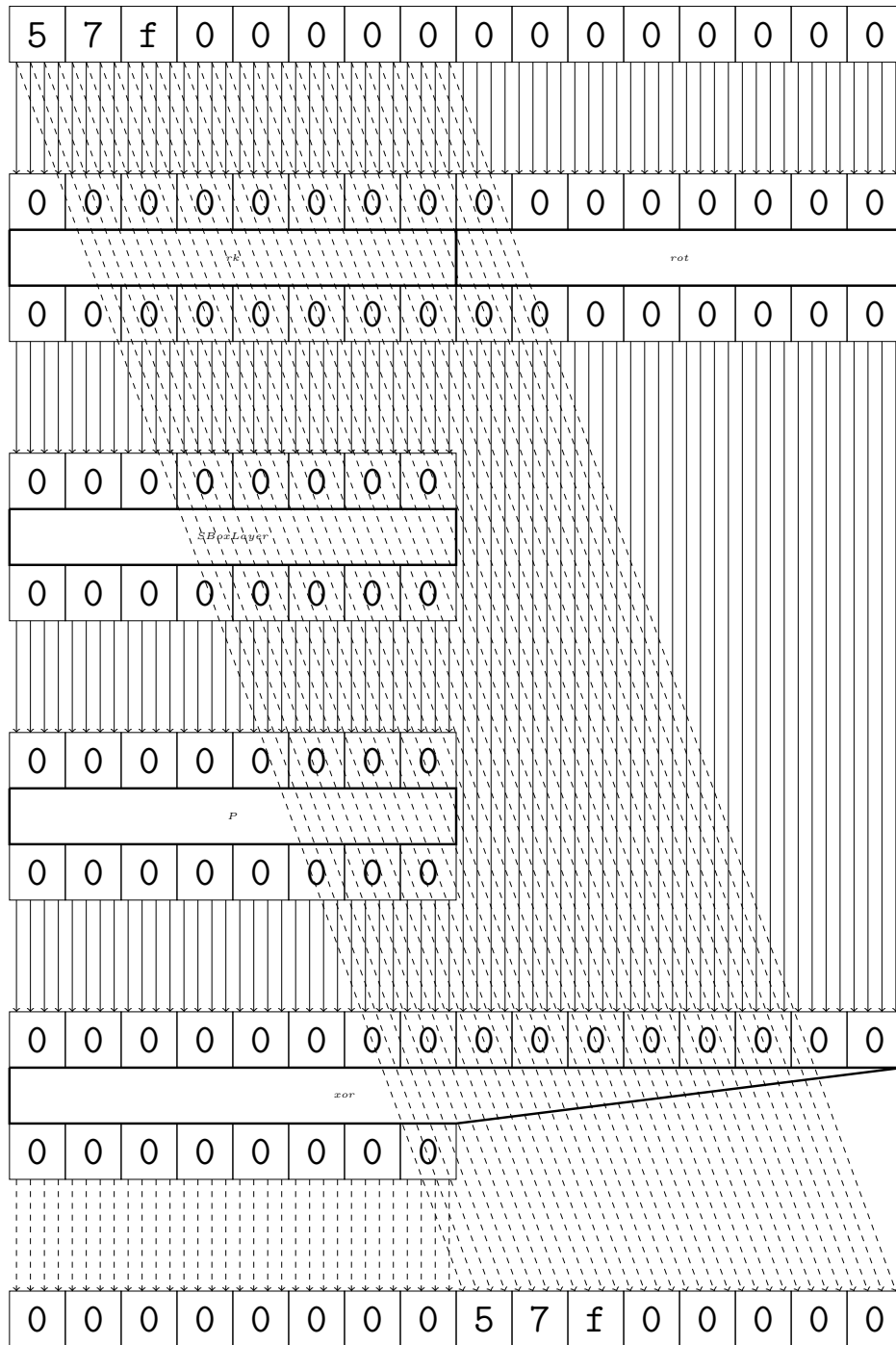
21 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



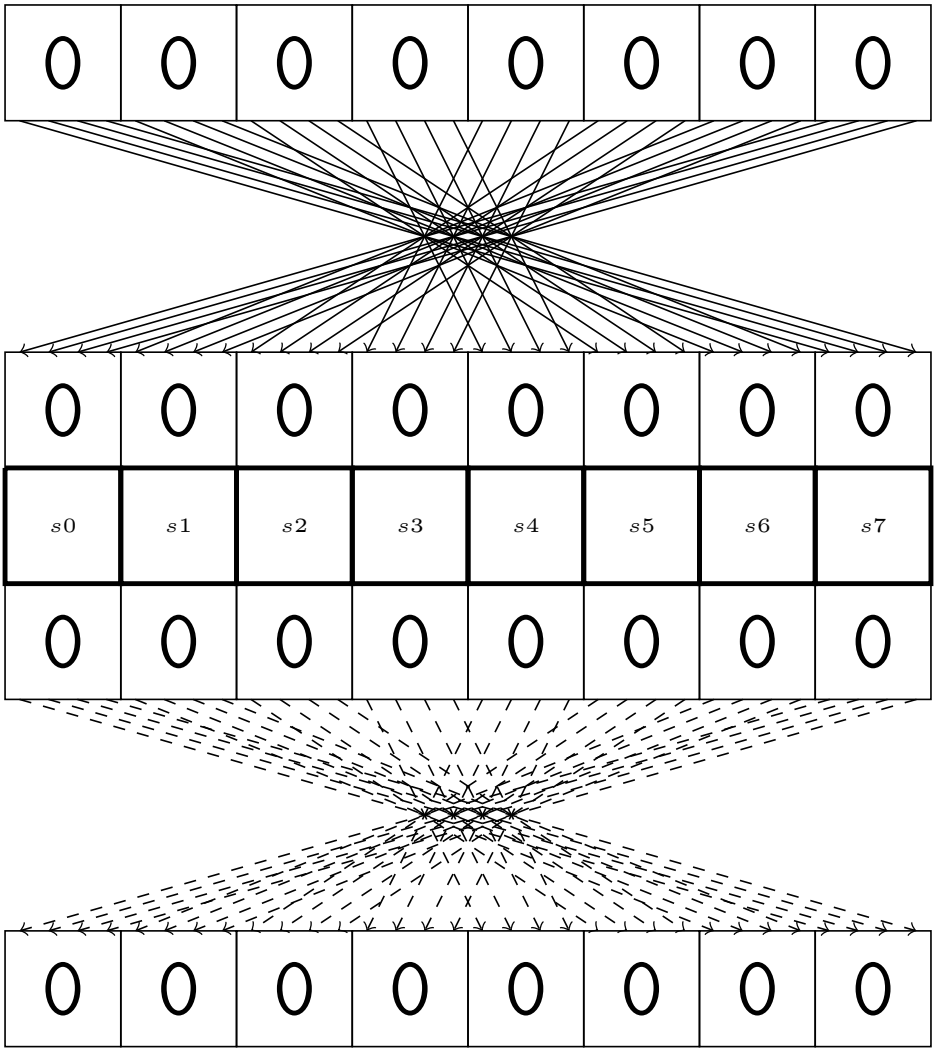
22 LBlock_round

Maximal linear correlation: $2^{-0.0}$



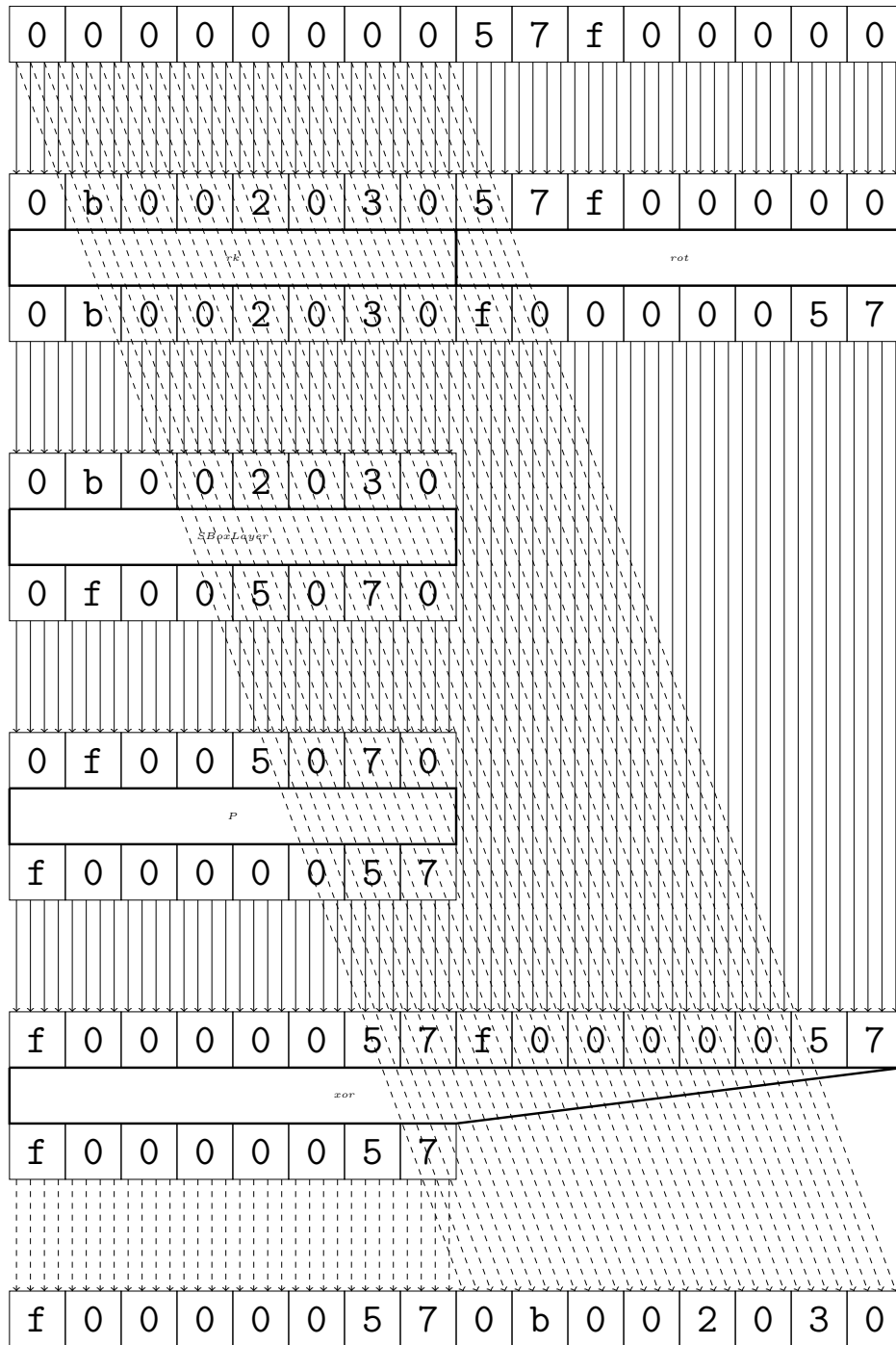
23 SBoxLayer

Maximal linear correlation: $2^{-0.0}$



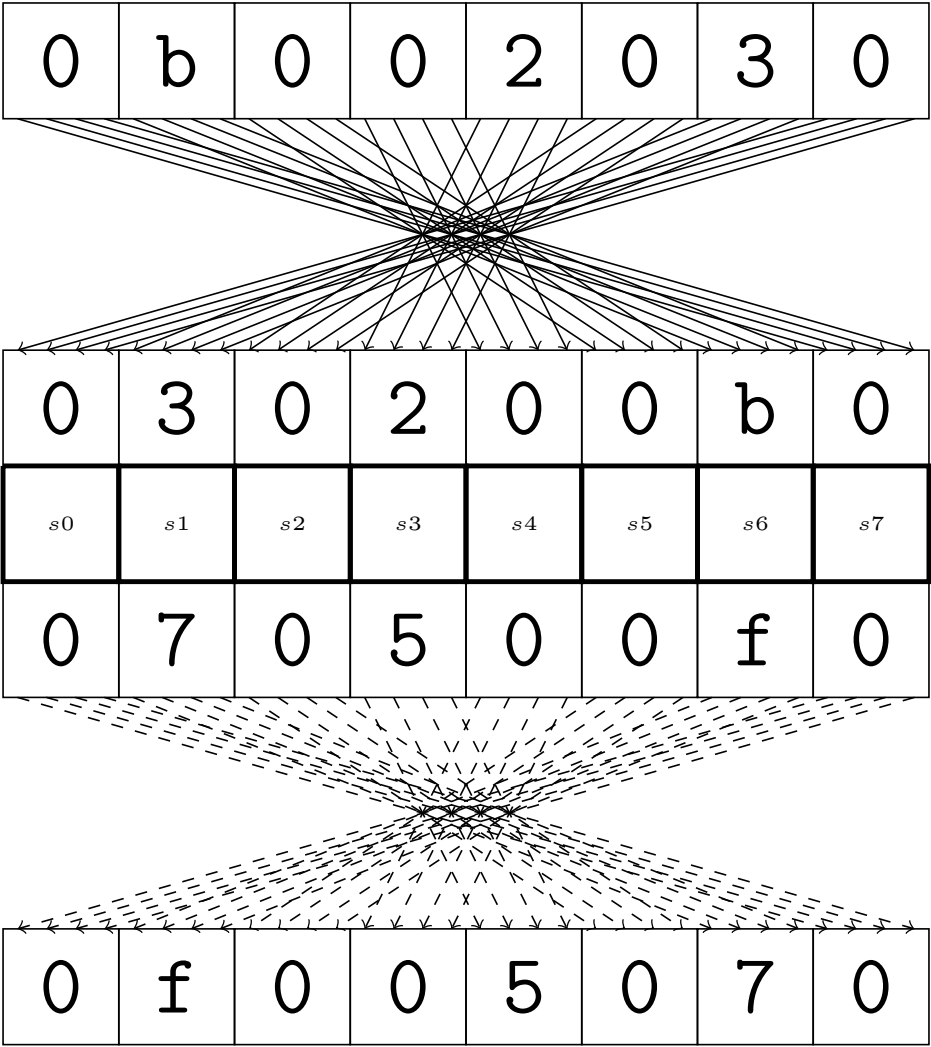
24 LBlock_round

Maximal linear correlation: $2^{-4.0}$

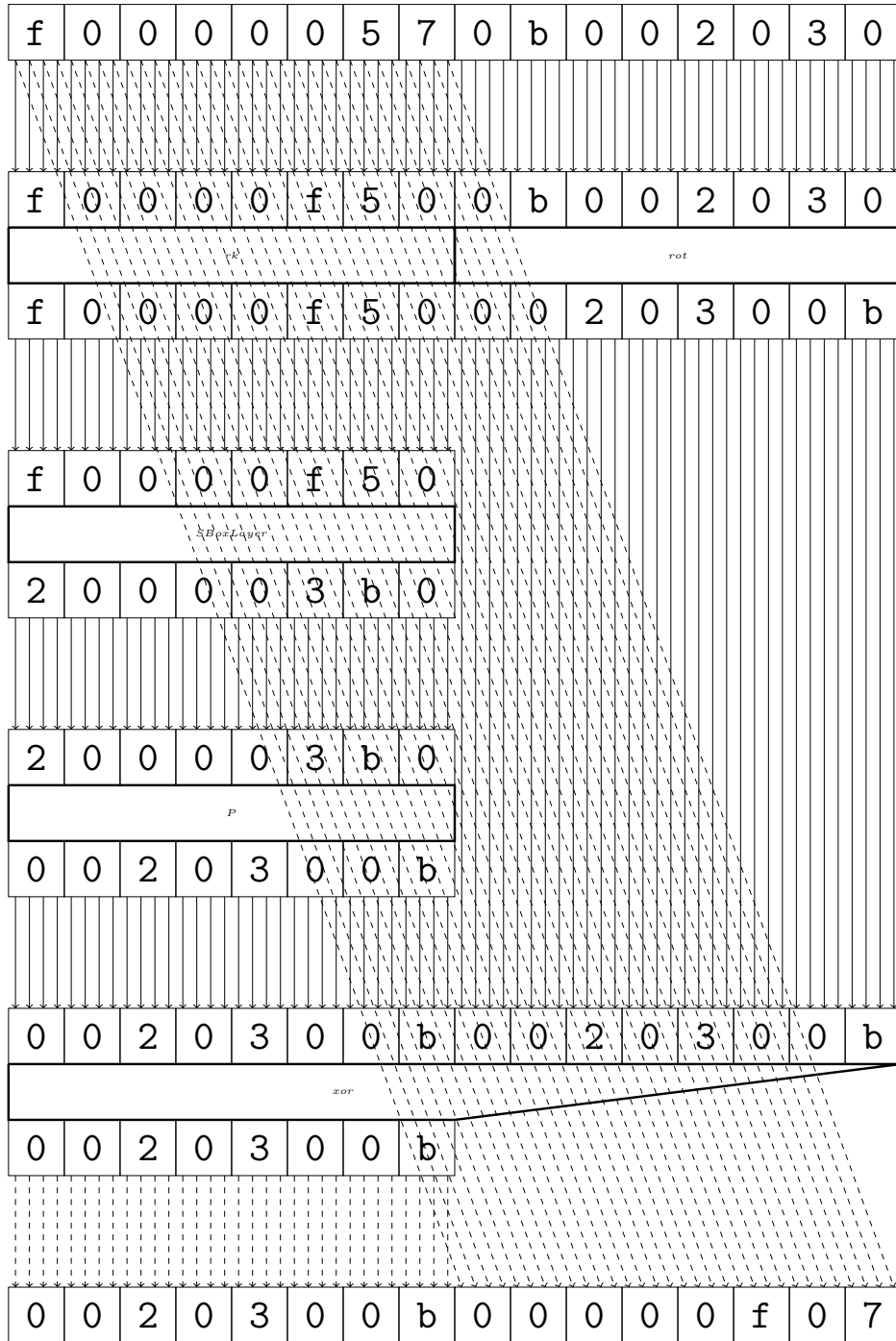


25 SBoxLayer

Maximal linear correlation: $2^{-4.0}$

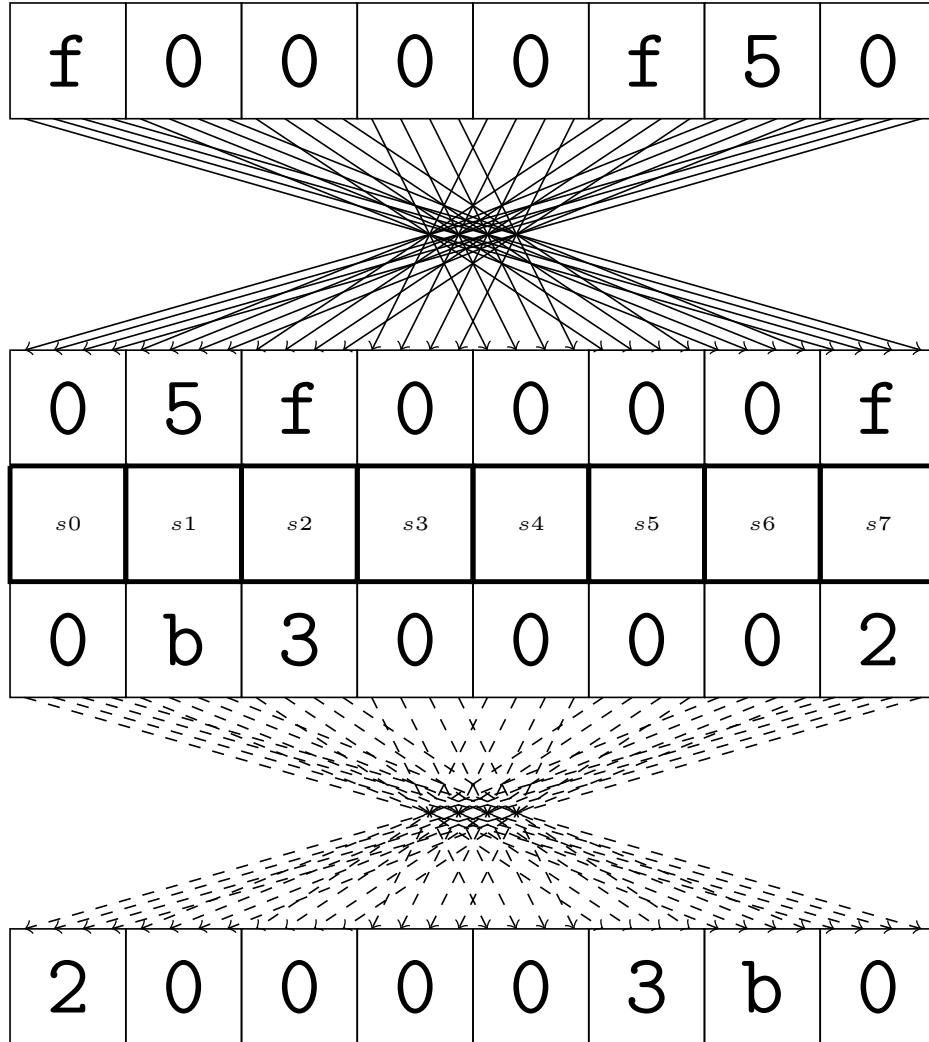


26 LBlock_round

Maximal linear correlation: $2^{-3.0}$ 

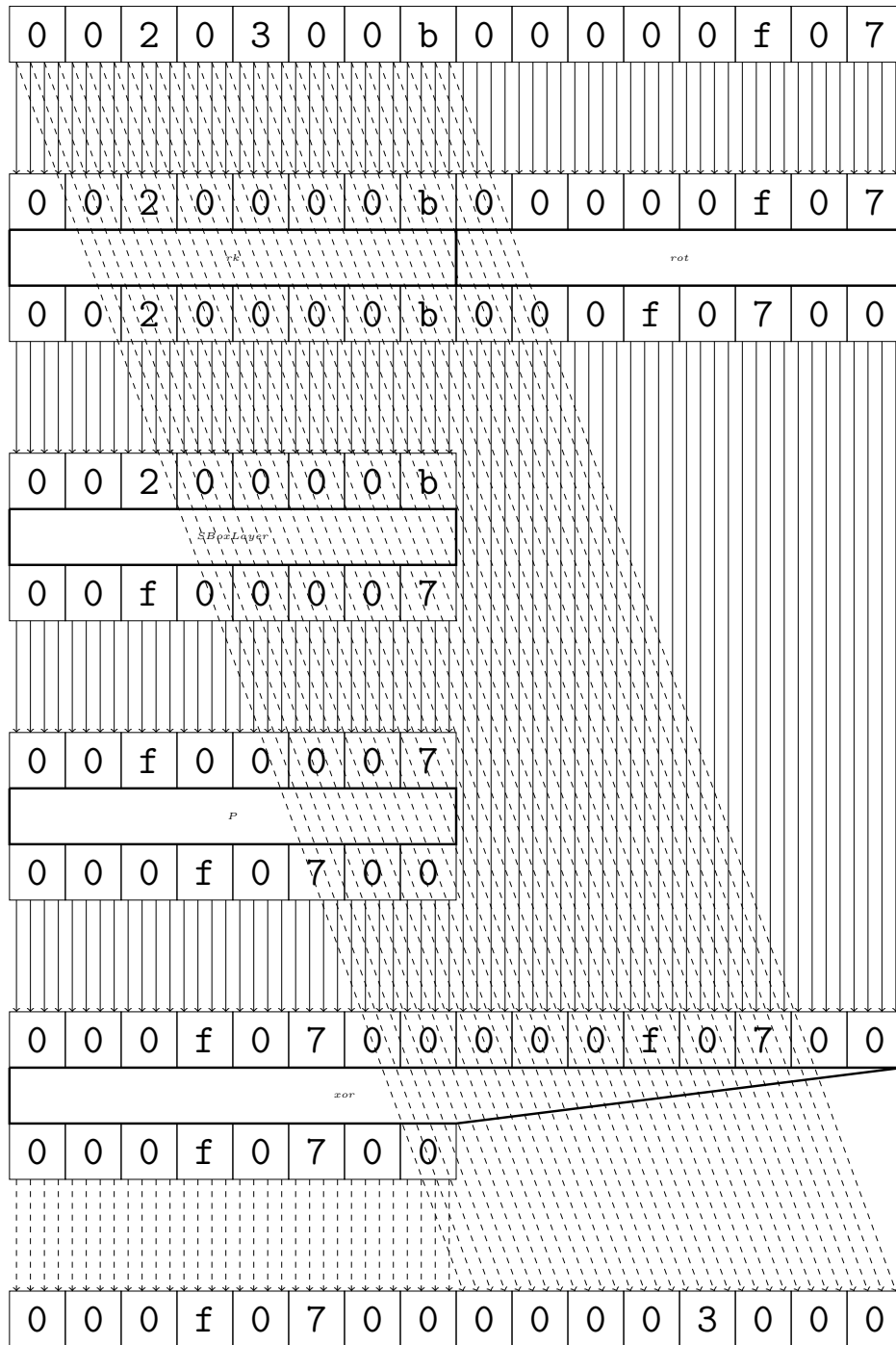
27 SBoxLayer

Maximal linear correlation: $2^{-3.0}$



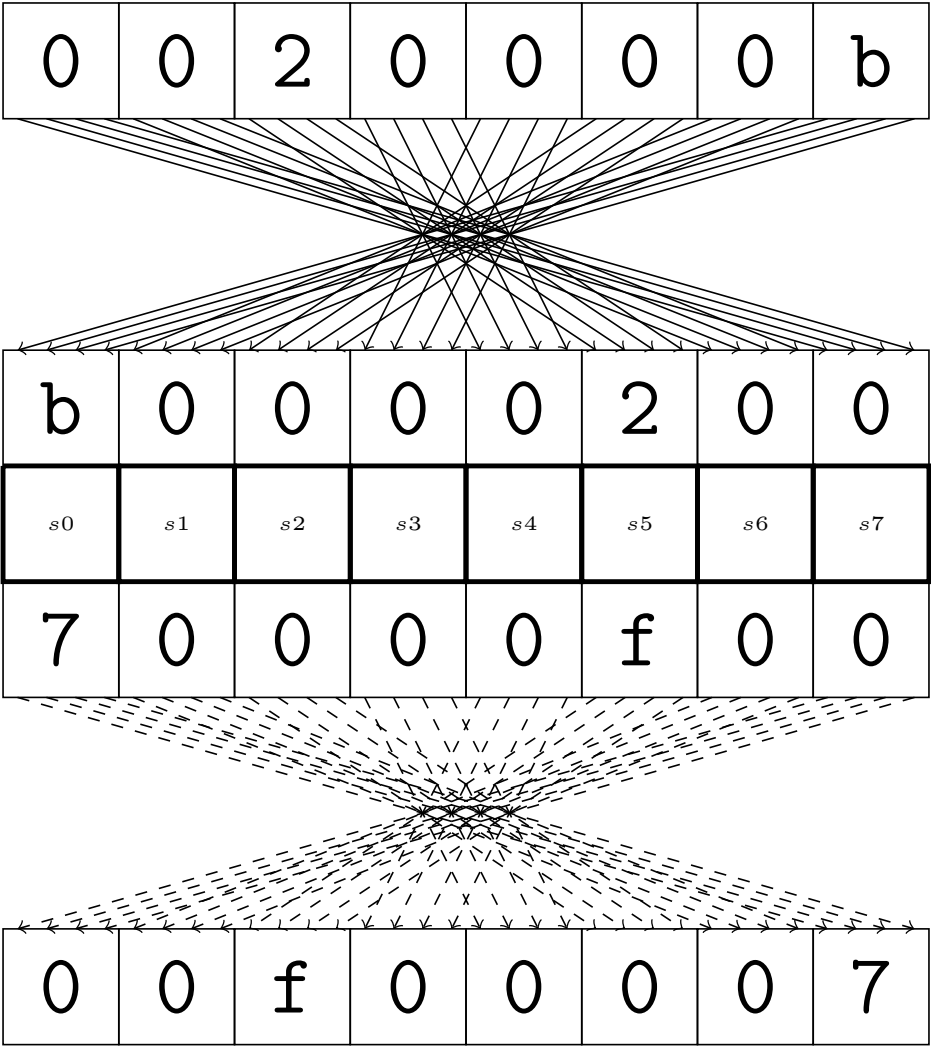
28 LBlock_round

Maximal linear correlation: $2^{-2.0}$

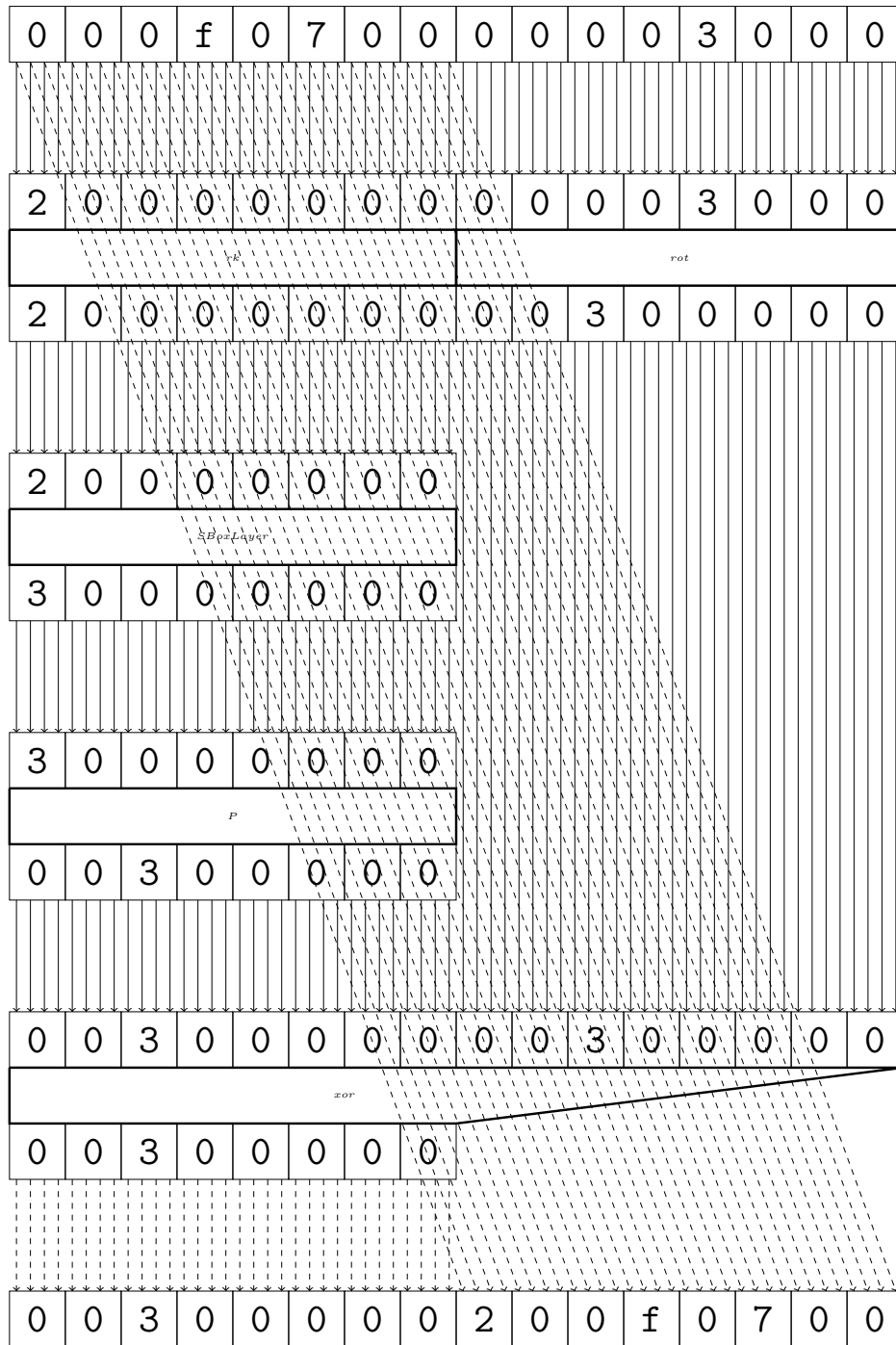


29 SBoxLayer

Maximal linear correlation: $2^{-2.0}$



30 LBlock_round

Maximal linear correlation: $2^{-1.0}$ 

31 SBoxLayer

Maximal linear correlation: $2^{-1.0}$

